Analysis and Training Evaluation of Nitroaromatic Explosives Detection Data Using Fluorescence and Machine Learning

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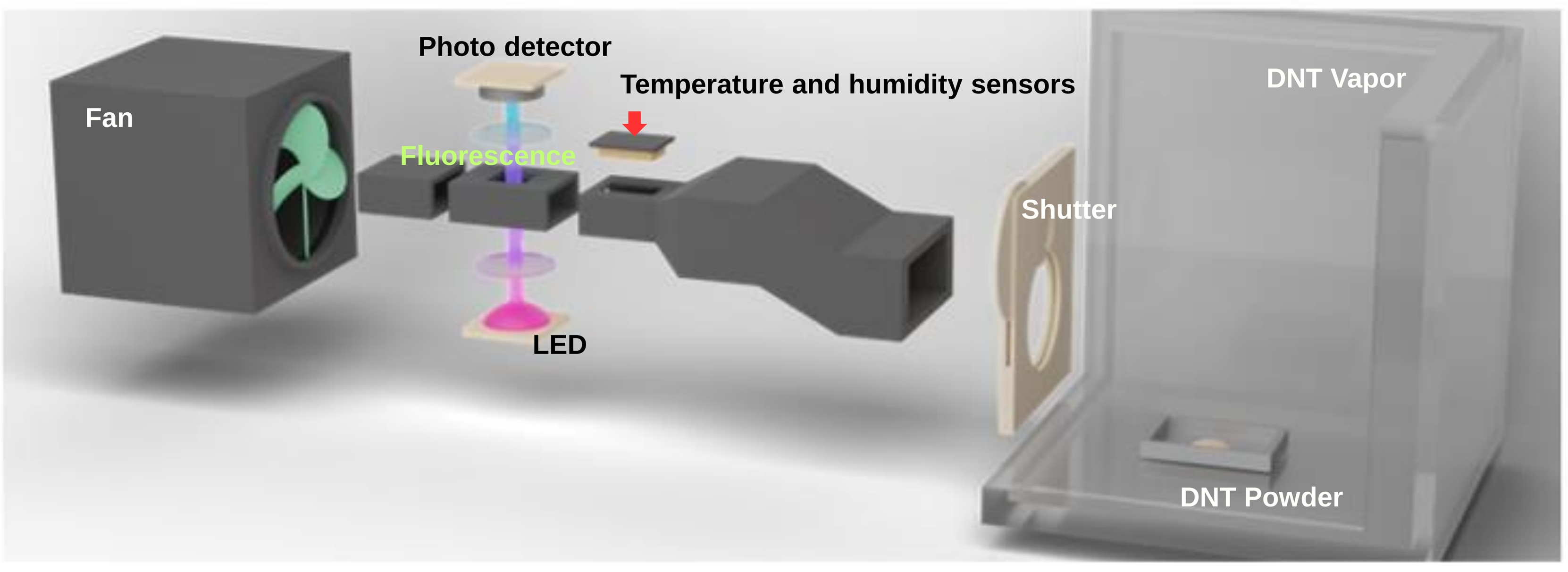
ABSTRACT

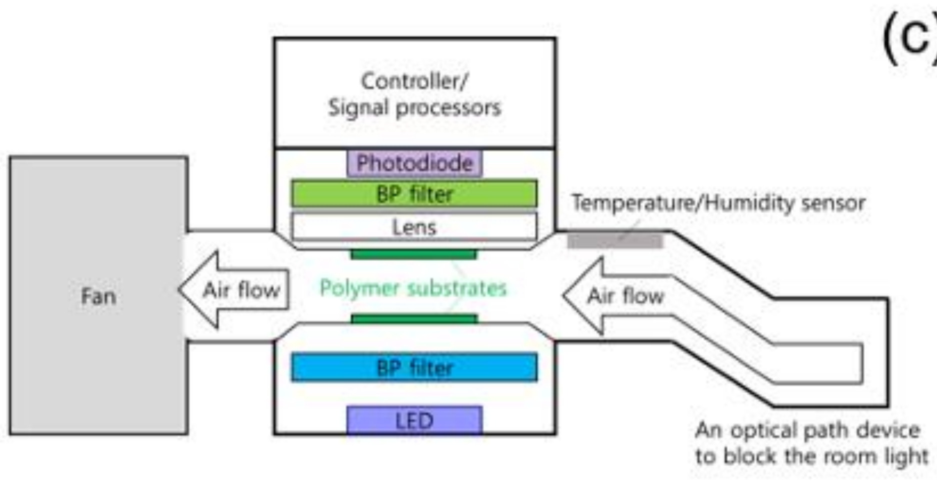
The detection of nitroaromatic explosives using fluorescence is significantly affected by environmental factors such as temperature, humidity, and vapor pressure. To overcome these challenges, machine learning techniques are employed to process and analyze fluorescence measurement data. Five models were developed and evaluated, with their performance assessed based on the presence or absence of skip connections, regularizers, batch normalization, and different types of initializers. The training loss was recorded at 171.98, and the validation loss at 118.03. It was observed that skip connections significantly improved the model's performance, while the other factors either had minimal or negative effects on the performance.

INTRODUCTION

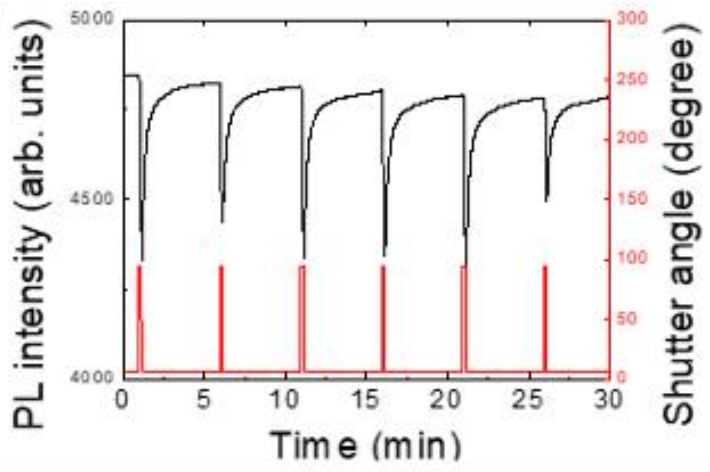
(a)



(b)



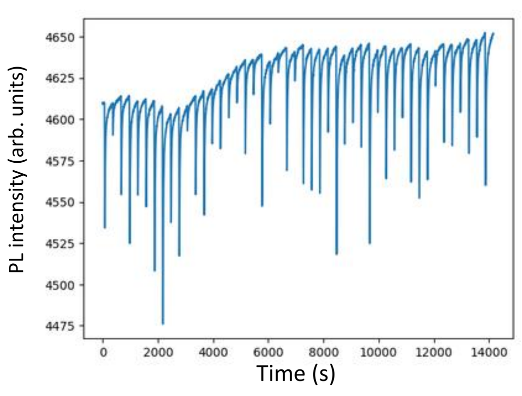
(c)

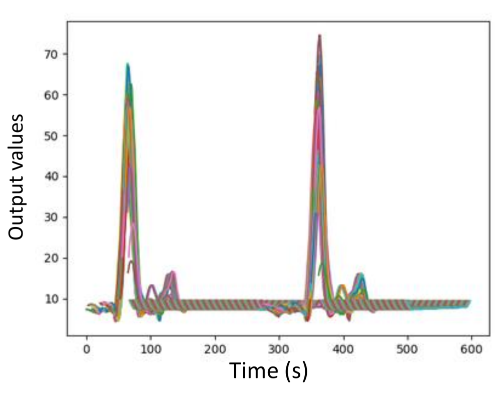


(a) A schematic picture of the experimental set-up consisting of a box with a shutter and a PL apparatus. 0.1 gram of 2,4-DNT powder was placed in a stainless-steel bowl. (b) A schematic diagram of the PL apparatus. (c) Photoluminescence response by shutter opening and closing. The period was 5 minutes and the shutter opening time was 10 seconds.

EXPERIMENTAL SETUP

- **Mean Squared Error (MSE) is a metric used to evaluate the performance of regression models by calculating the average of the squared differences between predicted and actual values. The loss refers to the model's error on the training dataset, with the goal of minimizing this value to improve the model's accuracy. Val_loss, or validation loss, measures the model's error on a separate validation dataset, which is used to monitor and prevent overfitting by assessing the model's generalization performance.**

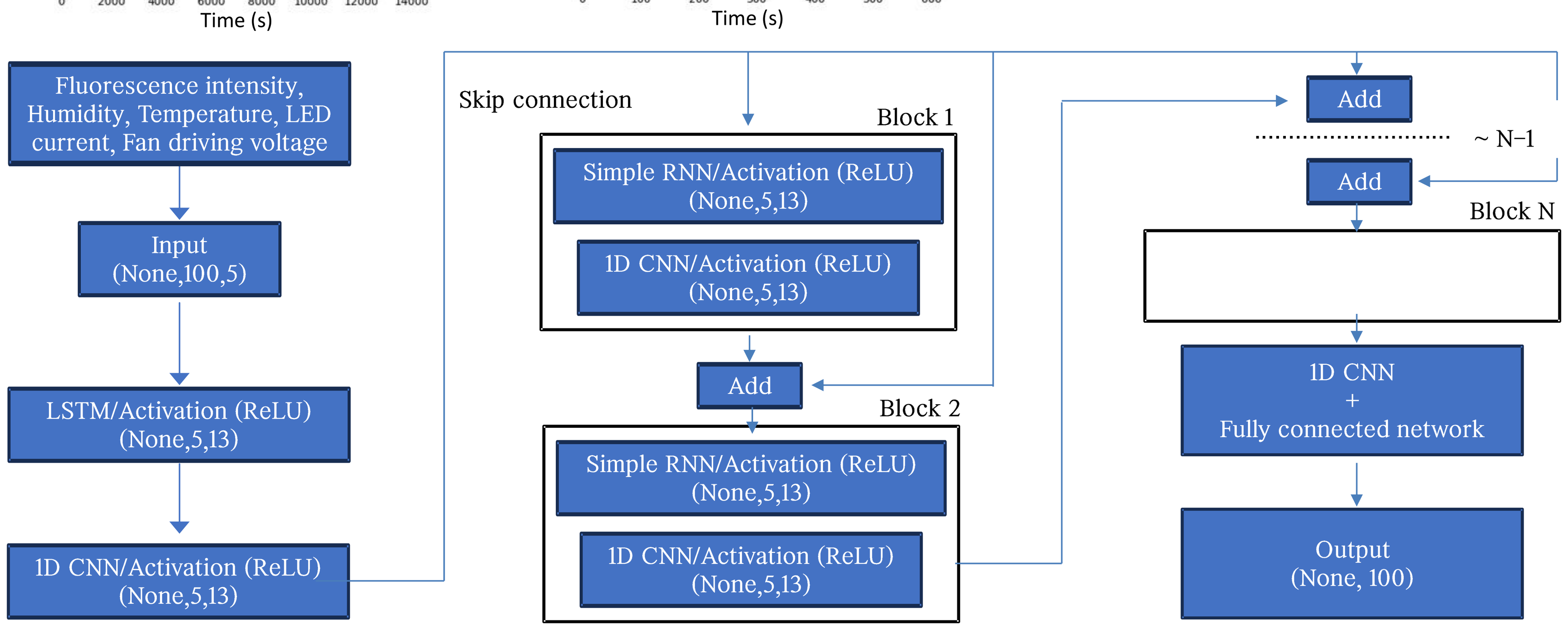




$$\mathrm{MSE} = \frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2$$

$$\mathrm{Loss} = \frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2$$

$$\mathrm{Val\_Loss} = \frac{1}{m}\sum_{i=1}^{m}(y_i - \hat{y}_i)^2$$



model 1
- N=4
- Skip connection
- Batch normalization
- Regularization (L1)
- Initializer = He normal

model 2
- N=4
- Skip connection
- No Batch normalization
- No Regularization
- Initializer = Glorot uniform (Keras default)

model 3
- N=4
- No Skip connection
- Batch normalization
- Regularization
- Initializer = He normal

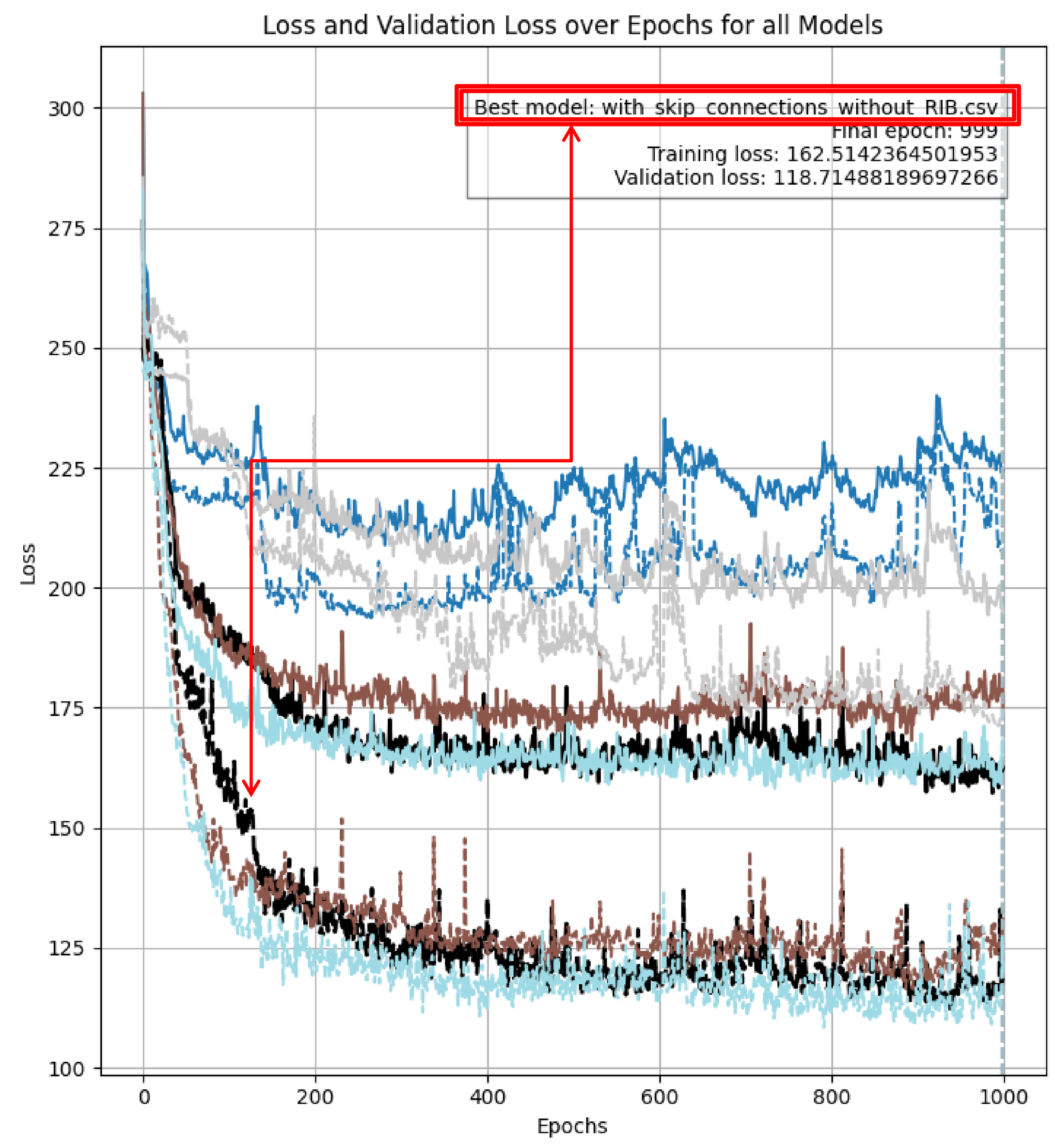
model 4
- N=4
- No Skip connection
- No Batch normalization
- No Regularization
- Initializer = Glorot uniform (Keras default)

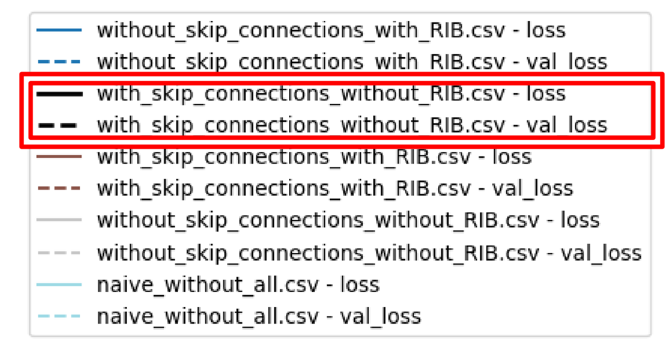
model 5
- N=1
- No Skip connection
- No Batch normalization
- No Regularization
- Initializer = Glorot uniform (Keras default)

REFERENCE

- **Noh, D.; Oh, E. Estimation of Environmental Effects and Response Time in Gas-Phase Explosives Detection Using Photoluminescence Quenching Method. Polymers 2024, 16, 908.**

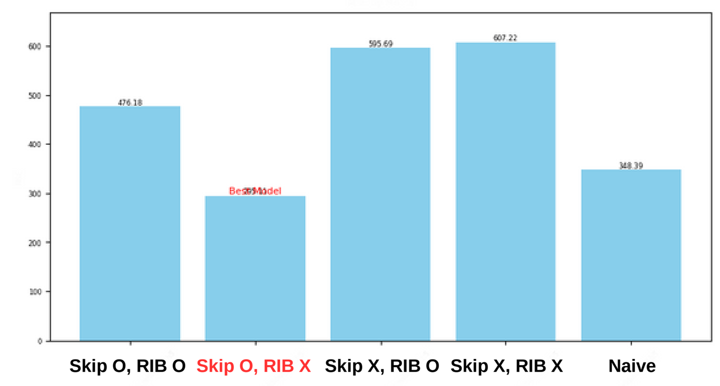
RESULTS

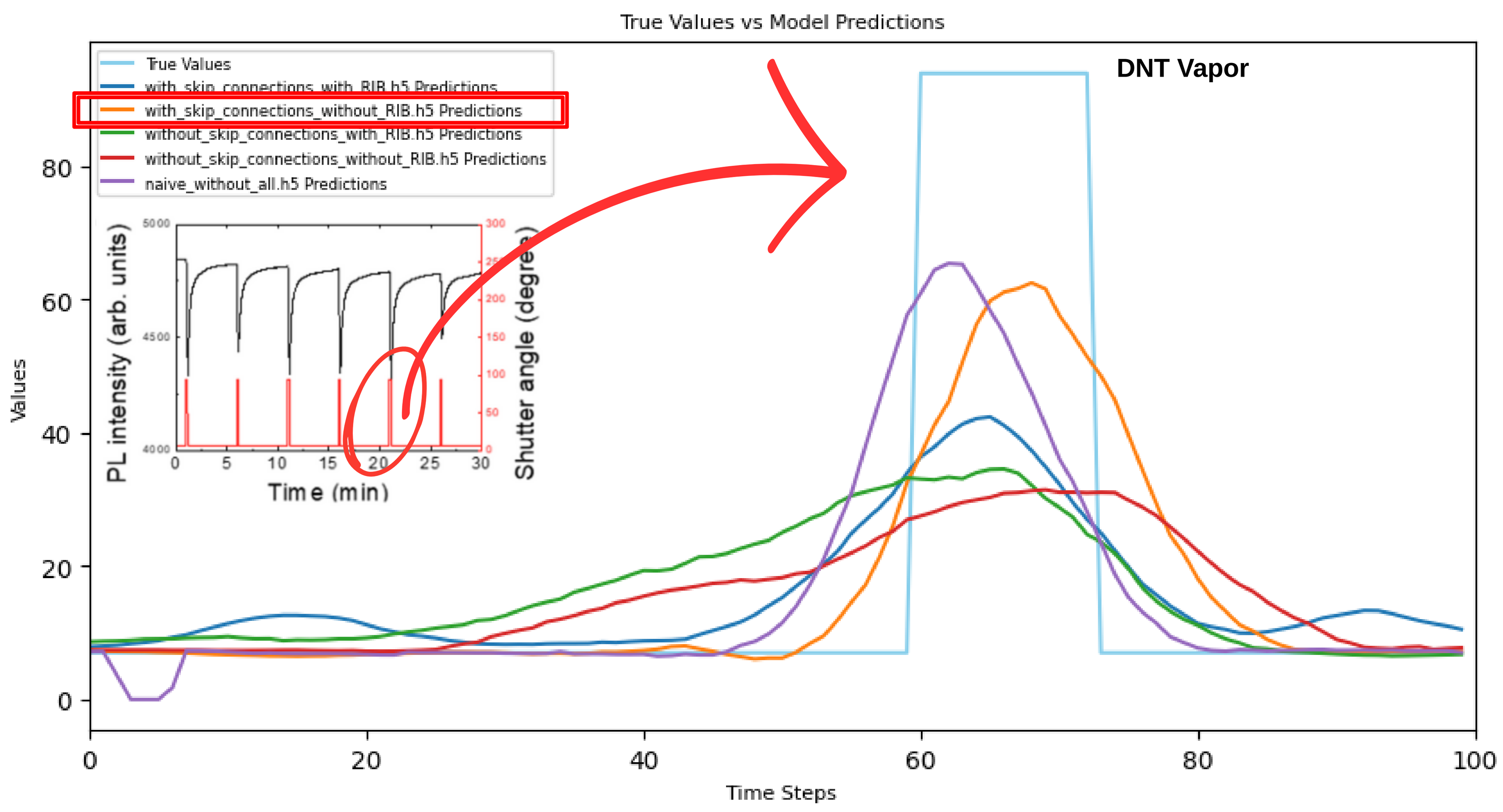




* Let's refer to Regularizers, Initializers, and Batch Normalization collectively as RIB.

MSE





- **Best Model: with_skip_connections_without_RIB [orange line] has the lowest MSE of 295.1118, indicating the best performance. This model's predictions closely follow the true values (light blue line), especially around the peaks and transitions.**
- **with_skip_connections_with_RIB model [green line] also shows relatively good fit but is less accurate than the best model. This suggests that skip connections significantly improve performance, and the presence of RIB may have a slight detrimental effect.**
- **Impact of statistical techniques: Models with RIB (with_skip_connections_with_RIB [green line] and without_skip_connections_with_RIB [blue line]) tend to deviate more from the true values compared to their counterparts without RIB.**
- **Effect of Skip Connections: The model without skip connections, without_skip_connections_without_RIB [red line], shows better fit compared to its RIB counterpart, but still lags behind models with skip connections.**
- **Naive Model: The naive_without_all model [purple line] is trained with only one run. Despite its minimal training iterations, it demonstrates relatively good performance with an MSE of 348.39. This highlights the efficiency of a simple model, suggesting that simpler approaches can sometimes yield better performance without overfitting."**

CONCLUSION

- **Skip connections: Skip connections in deep learning models prevent the vanishing gradient problem by bypassing intermediate layers, enabling stable training and resulting in lower loss values and enhanced performance.**
- **Statistical techniques (Regularizers, Initializers, Batch Normalization, RIB): Generally have a minimal or negative impact on model performance. The presence of RIB can slightly degrade performance, especially in models without skip connections.**
- **Optimal model: The best performance is achieved with models that include skip connections and exclude RIB elements.**